

Pronob Kumar Barman

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EDUCATION

University of Maryland, Baltimore County

PhD in Information Systems, CGPA: 3.85/4.00

01/2022 - 05/2025 (Anticipated)

Research Focus: Generative Models, NLP, Recommendation Systems, HCI

University of Kentucky

MS in Statistics, CGPA: 3.85/4.00

08/2019 - 05/2021

Advanced Theoretical Focus: Probability Theory, Statistical Inference, Experimental Design

University of Dhaka

MS in Statistics, CGPA: 3.46/4.00

10/2015 - 01/2017

BS in Statistics, CGPA: 3.46/4.00

01/2011 - 08/2015

PROFESSIONAL EXPERIENCE

University of Maryland, Baltimore County

Teaching Assistant

01/2022 – Present

- Enhanced graduate student outcomes in **ML and deep learning** projects, achieving **30% improvement in understanding**.

Technuf LLC

Data Scientist, Intern

06/2024 – 08/2024

- Assessed **economic viability** for Aphelia Student Connect, identifying revenue streams and reducing breakeven by **30%**.
- Integrated **Generative AI** into **SIEM/SOC systems**, automating threat detection and cutting response time by **40%**.

Data Scientist, Intern

08/2022 – 12/2022

- Optimized **SQL queries in PostgreSQL** for **200K+ records**, enhancing efficiency for NY Medicare/Medicaid.
- Designed **PowerBI dashboards** and executed **Flutter automation testing**, enhancing issue resolution.

University of Kentucky

Teaching Assistant

08/2019 – 05/2021

- Enhanced student comprehension in **statistical software and database languages** through applied examples.

Bangladesh Bank

Senior Data Scientist

04/2018 – 12/2021

- Developed and deployed **machine learning models** that enhanced **credit risk assessment** accuracy by 20%, leading to a 15% reduction in **default rates**.
- Utilized **statistical methods and machine learning models**, resulting in a 15% increase in **FDI inflows**, equating to an additional \$500 million annually.

LEADERSHIP & AWARDS

- General Secretary** of the Statistics Student Association, two terms.
- Most Consistent Intern of the Month** at Technuf LLC (Aug 2024).

SKILLS

- Languages:** Python, R, SAS, Stata, SQL, Java, Matlab
- Generative AI:** LangChain, OpenAI API, Hugging Face
- Tools:** TensorFlow, PyTorch, Tableau, NVivo, Hadoop, AWS
- Soft Skills:** Leadership, Communication, Problem-Solving

PHD RESEARCH PROJECT

Online Healthcare Support Group Formation and Recommendation System

- Developed an automated support system using **generative models** and **topic modeling**, enhancing **coherence by 60%** and improving **personalized suggestions by 75%**; released an [R package for gSTM](#).
- Conducted a **user experience survey via Qualtrics** to evaluate benefits, features, and privacy, directing platform enhancements based on user feedback.

OTHER PROJECTS

LLM Generative AI with LangChain and OpenAI API [Github](#)

- Developed a **Generative AI system** with **Pinecone** and **ChromaDB**, enabling advanced retrieval-augmented semantic search.

End-to-End Medical Chatbot with Llama2 and Hugging Face API [Github](#)

- Developed a **real-time medical chatbot with Llama2**, improving **patient interaction flow by 30%** through **LangChain** enhancements.